

IT637 Introduction to Algorithms (Winter 2026)

Instructor: Amit Mankodi, 4111, Faculty Block-4

Course Placement: MSc. IT, Semester-II

Course format: 3 Lecture hours and 2 Lab hours.

Course content: This course intends to provide a rigorous introduction to fundamental techniques in the design and analysis of algorithms. The course aims to cover the asymptotic notation, divide and conquer techniques, solving recurrences, greedy algorithms, dynamic programming, sorting and searching algorithms, depth-first search, breadth-first search, topological sort, minimum spanning trees, shortest path problem, pattern matching, NP-completeness, and approximation algorithms.

Text Book:

Introduction to Algorithms, 2nd edition, by Cormen, Leiserson, Rivest, Stein (CLRS)

Evaluation plan:

1. Insem1 exam: 15%
2. Insem2 exam: 20%
3. Endsem exam: 30
4. Lab Evaluation: 20%
5. Pop-up quizzes: 10%
6. Attendance: 5%

Course Outcomes:

After successful completion of the course:

- Students will learn various approaches, such as recursion, dynamic programming, backtracking, etc, to designing an algorithm.
- Students will learn searching and sorting algorithms.
- Students will learn advanced data structures such as trees and graphs and their respective algorithms.
- Students will be able to perform comparative analysis of algorithms using the time and space complexity.
- Students will be able to apply logical thinking to solve problems by applying designed algorithms

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X						X			X

Lecture Schedule:

Sl. No.	Description	No. of Lectures
1	Asymptotic Notation: Algorithm analysis in time and space complexity, Big Oh, Big Omega and Big Theta notations	6

2	Algorithm Design Technique: recurrence equation, solving recurrence, iterative method, tree method, master method	6
3	Searching and sorting algorithms: bubble sort, selection sort, insertion sort, merge sort, quicksort	4
4	Binary Search Tree: inorder, preorder, postorder tree traversal, BST operations: insert, delete	4
5	Graph Search Algorithms: breadth-first search, depth-first search	2
6	Graph Algorithms: strongly connected components	2
7	Graph Algorithms: Shortest Paths Algorithms: Bellman-Ford, single source shortest path, Dijkstra, minimum spanning tree	8
8	Tree algorithms: AVL tree	4
9	String Matching Algorithms: Rabin-Karp, Finite automaton, Knuth-Morris-Pratt	2
10	Class of Algorithms: P, NP and NP Complete algorithms	2